Active Maintenance Report: coppock_etal_2017

Coppock, Gerber, Green and Kern (2017, Political Analysis): Combining Double Sampling and Bounds to Address Nonignorable Missing Outcomes in Randomized Experiments

2026-08-01

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Drafted by Claude Opus 5 under the supervision of Alex Coppock.

This repository holds the actively maintained replication code for Coppock, Gerber, Green and Kern (2017), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/pan.2016.6
Replication archive	10.7910/DVN/AQB4MP
Preanalysis plan	osf.io/wkj8v

The data are not redistributed here. The deposit is 231 KB across seven files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_etal_2017.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its seven files, installs the `attrition` package from the deposited source tarball if it is not already present, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, estimatr, modelsummary, knitr, kableExtra, gridExtra, here, and `attrition`, which `install_attrition.R` handles. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about seven seconds. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.** The three PDF figures differ on every run because a PDF records the time it was written; the CSV, TeX and PNG output comes back byte-identical.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Not without intervention, and the obstacle is a bug in the deposit rather than anything that has aged. `experimental_analysis_main_text.R` prints Tables 1, 2 and 3 and then stops at a line that subtracts rows from an object called `mturk_results`, which the script never creates. The object it means is `table_3`, assembled four lines earlier under a different name. Everything after that line, which is the 65 percent confidence interval reduction the results section quotes and the entire sensitivity analysis behind Figure 2, is unreachable. That leaves 2 of the 130 recorded claims with no archive value at all, and both of them are numbers the article prints.

The other two scripts run. `simulation_analysis.R` produces Figure 1 and Appendix Figure 1 with two deprecation warnings and no errors. `experimental_analysis_appendix.R` runs and reproduces Appendix Table 1 exactly, but only against the `commarobust` package as deposited. The current version of `commarobust` on GitHub also calls itself 0.1.0 and exports nothing at all: it has been reduced to a message telling the user to install `estimatr` instead. A reader who follows the archive's own README, which recommends `devtools::install_github()` over the bundled tarball, gets a package that fails on the first line that uses it, and `packageVersion()` reports the same number either way.

The remaining 128 claims all reproduce, including every cell of all five published tables.

1.2 Does the maintained rewrite reproduce the paper?

Yes, without exception. All 130 recorded claims match the published values to reported precision, and the two the archive cannot reach are among them: replacing the stale object name gives a 65.0 percent reduction in confidence interval width where the paper says 65 percent, and the sensitivity analysis gives a δ^* of 0.0707 where the paper says 0.07.

2 Paper overview

Citation: Coppock, A., Gerber, A. S., Green, D. P. and Kern, H. L. (2017). "Combining double sampling and bounds to address nonignorable missing outcomes in randomized experiments." *Political Analysis*, 25(2), 188-206. DOI: 10.1017/pan.2016.6

Summary: Missing outcomes break the guarantee randomization provides, and the usual repairs assume that missingness is ignorable. The paper proposes an alternative that does not: combine a double sampling design, in which a random subset of the initial nonrespondents is pursued with a larger incentive, with nonparametric worst-case bounds on the outcomes still missing after that second effort. It derives the bounds estimator under double sampling, gives analytic variances and confidence intervals, adds poststratification for covariate adjustment, and adds a sensitivity analysis indexed by the fraction δ of remaining nonrespondents for whom ignorability fails. Monte Carlo simulations show how the identification region and the

confidence interval respond to the two response rates. The application is a two-wave Mechanical Turk replication of Levendusky and Malhotra (2016) in which 1,980 subjects read either a polarized or a moderate news story, 1,444 responded in the initial Wave 2 sample, and 50 nonrespondents per arm were offered four times the original incentive, of whom 72 complied. Worst-case bounds from the initial sample alone are $(-1.54, 1.71)$ and say nothing. Double sampling with poststratification narrows the associated confidence interval from 3.50 to 1.23, a 65 percent reduction, while the sensitivity analysis shows the significant naive estimate collapses once ignorability fails for even 7 percent of the remaining nonrespondents.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Status on current R	Resolution
experimental_analysis_main_text.R	Prints Tables 1, 2 and 3, then stops: object 'mturk_results' not found	The object meant is table_3, built four lines above under another name
experimental_analysis_appendix.R	Runs against the deposited commarobust; fails against the current one	install.packages('original/commarobust_0.1.0.tar.gz', repos = NULL, type = 'source')
simulation_analysis.R	Runs; two deprecation warnings	guide = FALSE becomes guide = 'none'; %+% becomes +

The deposit is unusually well prepared in one respect and unusually exposed in another. It bundles the source tarballs of both packages the analysis depends on, `attrition` and `commarobust`, neither of which has ever been on CRAN. That is the right instinct, and it is what makes the appendix script reproducible today. The README then undercuts it: it recommends installing from GitHub instead, calling that route "easy," and offers the tarballs only "in the interest of full replicability." Ten years on, the easy route is the broken one.

3.1 The two packages

`attrition` implements the paper's method and is still on GitHub. The deposited version is 0.0.0.9000; the current one is 0.1.0. Both were installed and run against every quantity in this repository, and they agree to twelve significant figures. `install_attrition.R` installs the deposited tarball if `attrition` is absent, because the tarball's checksum is pinned in `original_manifest.csv` and a GitHub repository's contents are not.

`commarobust` is a different case. It was the author's prototype for robust standard errors around `lm()`, and it was superseded by `estimatr`, whose `lm_robust()` defaults to the same HC2 estimator. The current GitHub `commarobust` has been emptied out to a startup message pointing at `estimatr`, and it exports no functions. It nevertheless still calls itself version 0.1.0, exactly like the deposited tarball, which exports six. Two packages with the same name and

the same version number differ in whether the archive runs, and nothing in `sessionInfo()` would tell a reader which one they have.

3.2 The stale object name

The main text script builds Table 3 and then reaches for it under a name it does not have:

```
table_3 <- cbind(no_ds = cis_1, ds = cis_2, ds_ps = cis_3)
# ... four lines of row labels and a print ...
widths <- mturk_results[2,] - mturk_results[1,]
(widths[1] - widths[3]) / widths[1]
```

`mturk_results` is never assigned anywhere in the archive. The name presumably survived an interactive session in which the object had been called that, and the rename was not carried through before the file was deposited. This is the class of defect that only a clean-session run finds: in the author's own session the old object was still in the workspace, so the script worked.

The cost is not cosmetic. The line that fails computes the 65 percent reduction in confidence interval width, which is the sentence the results section builds toward, and everything below it, including the entire sensitivity analysis and Figure 2, never executes. Both values are correct once the name is fixed, so the archive contains a correct analysis that it cannot run.

4 Errata

The maintained rewrite makes one correction to the archive's code: `mturk_results` becomes the object that was meant. Nothing else in the estimator, the specification or the sample is changed.

Table 2: The two published claims the deposited script cannot reach, and what the rewrite produces.

Claim	Published	Archive	Rewrite
Section 4.2 reduction in 95% CI width	65 percent	not computed, script stops	0.6503
Section 4.3 sensitivity parameter	0.07	not computed, script stops	0.0707

No published value changes. The correction recovers two numbers the archive was unable to produce; both agree with the article.

5 Number-by-number comparison

Table 3: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means the script stops before reaching the quantity.

Location	Comparison	DV	Quantity	Paper	Archive	Match
table_1	Moderate		N_initial	995	995	1
table_1	Moderate		responded	731	731	1
table_1	Moderate		did_not_respond	264	264	1
table_1	Moderate		invited	50	50	1
table_1	Moderate		responded_to_invitation	39	39	1
table_1	Polarized		N_initial	985	985	1
table_1	Polarized		responded	713	713	1
table_1	Polarized		did_not_respond	272	272	1
table_1	Polarized		invited	50	50	1
table_1	Polarized		responded_to_invitation	33	33	1
table_1	Total		N_initial	1980	1980	1
table_1	Total		responded	1444	1444	1
table_1	Total		did_not_respond	536	536	1
table_1	Total		invited	100	100	1
table_1	Total		responded_to_invitation	72	72	1
table_2	Moderate	PP	IR_mean	3.542	3.542	1
table_2	Moderate	PP	IR_sd	1.243	1.243	1
table_2	Moderate	PP	IR_N	731	731	1
table_2	Moderate	PP	DS_mean	3.583	3.583	1
table_2	Moderate	PP	DS_sd	1.367	1.367	1
table_2	Moderate	PP	DS_N	39	39	1
table_2	Polarized	PP	IR_mean	3.668	3.668	1
table_2	Polarized	PP	IR_sd	1.27	1.27	1
table_2	Polarized	PP	IR_N	713	713	1
table_2	Polarized	PP	DS_mean	3.826	3.826	1
table_2	Polarized	PP	DS_sd	1.313	1.313	1
table_2	Polarized	PP	DS_N	33	33	1
table_3	pol_vs_mod	PP	no_ds_ci_lower	-1.6691	-1.6691	1
table_3	pol_vs_mod	PP	no_ds_ci_upper	1.8359	1.8359	1
table_3	pol_vs_mod	PP	no_ds_wcb_low	-1.5391	-1.5391	1
table_3	pol_vs_mod	PP	no_ds_wcb_high	1.7097	1.7097	1
table_3	pol_vs_mod	PP	no_ds_var_low	0.0062	0.0062	1
table_3	pol_vs_mod	PP	no_ds_var_high	0.0059	0.0059	1
table_3	pol_vs_mod	PP	ds_ci_lower	-0.5283	-0.5283	1
table_3	pol_vs_mod	PP	ds_ci_upper	0.7452	0.7452	1
table_3	pol_vs_mod	PP	ds_wcb_low	-0.3417	-0.3417	1
table_3	pol_vs_mod	PP	ds_wcb_high	0.5718	0.5718	1
table_3	pol_vs_mod	PP	ds_var_low	0.0129	0.0129	1
table_3	pol_vs_mod	PP	ds_var_high	0.0111	0.0111	1
table_3	pol_vs_mod	PP	ds_ps_ci_lower	-0.529	-0.529	1
table_3	pol_vs_mod	PP	ds_ps_ci_upper	0.6966	0.6966	1
table_3	pol_vs_mod	PP	ds_ps_wcb_low	-0.3444	-0.3444	1
table_3	pol_vs_mod	PP	ds_ps_wcb_high	0.5257	0.5257	1
table_3	pol_vs_mod	PP	ds_ps_var_low	0.0126	0.0126	1
table_3	pol_vs_mod	PP	ds_ps_var_high	0.0108	0.0108	1
text	pol_vs_mod	PP	naive_ate_coef	0.126	0.126	1
text	pol_vs_mod	PP	naive_ate_se	0.066	0.066	1
text	pol_vs_mod	PP	naive_ate_pval	0.057	0.057	1
text	pol_vs_mod	PP	ci_width_no_ds	3.5	3.505	1
text	pol_vs_mod	PP	ci_width_ds_ps	1.23	1.2256	1
text	pol_vs_mod	PP	reduction_pct	0.65		0
text	pol_vs_mod	PP	delta_star	0.07		0
table_a1	Placebo	PP	wave1_coef	0.352	0.352	1
table_a1	Placebo	PP	wave1_se	0.067	0.067	1
table_a1	Placebo	PP	wave2_coef	0.042	0.042	1
table_a1	Placebo	PP	wave2_se	0.066	0.066	1
table_a1	Placebo	EX	wave1_coef	0.019	0.019	1
table_a1	Placebo	EX	wave1_se	0.035	0.035	1
table_a1	Placebo	EX	wave2_coef	-0.011	-0.011	1
table_a1	Placebo	EX	wave2_se	0.034	0.034	1
table_a1	Polarized	PP	wave1_coef	0.468	0.468	1

Table 3: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means the script stops before reaching the quantity. *(continued)*

Location	Comparison	DV	Quantity	Paper	Archive	Match
table_a1	Polarized	PP	wave1_se	0.067	0.067	1
table_a1	Polarized	PP	wave2_coef	0.126	0.126	1
table_a1	Polarized	PP	wave2_se	0.066	0.066	1
table_a1	Polarized	EX	wave1_coef	-0.031	-0.031	1
table_a1	Polarized	EX	wave1_se	0.034	0.034	1
table_a1	Polarized	EX	wave2_coef	-0.03	-0.03	1
table_a1	Polarized	EX	wave2_se	0.034	0.034	1
table_a1	Constant	PP	wave1_coef	3.108	3.108	1
table_a1	Constant	PP	wave1_se	0.046	0.046	1
table_a1	Constant	PP	wave2_coef	3.542	3.542	1
table_a1	Constant	PP	wave2_se	0.046	0.046	1
table_a1	Constant	EX	wave1_coef	1.531	1.531	1
table_a1	Constant	EX	wave1_se	0.024	0.024	1
table_a1	Constant	EX	wave2_coef	1.617	1.617	1
table_a1	Constant	EX	wave2_se	0.024	0.024	1
table_a1	all	PP	wave1_N	2156	2156	1
table_a1	all	PP	wave2_N	2156	2156	1
table_a1	all	EX	wave1_N	2156	2156	1
table_a1	all	EX	wave2_N	2156	2156	1
table_a1	all	PP	wave1_R2	0.024	0.024	1
table_a1	all	PP	wave2_R2	0.002	0.002	1
table_a1	all	EX	wave1_R2	0.001	0.001	1
table_a1	all	EX	wave2_R2	4e-04	4e-04	1
table_a2	pol_vs_mod	PP	ci_lower	-0.5283	-0.5283	1
table_a2	pol_vs_mod	PP	ci_upper	0.7452	0.7452	1
table_a2	pol_vs_mod	PP	wcb_low	-0.3417	-0.3417	1
table_a2	pol_vs_mod	PP	wcb_high	0.5718	0.5718	1
table_a2	pol_vs_mod	PP	var_low	0.0129	0.0129	1
table_a2	pol_vs_mod	PP	var_high	0.0111	0.0111	1
table_a2	pol_vs_mod	EX	ci_lower	-0.6087	-0.6087	1
table_a2	pol_vs_mod	EX	ci_upper	0.6032	0.6032	1
table_a2	pol_vs_mod	EX	wcb_low	-0.471	-0.471	1
table_a2	pol_vs_mod	EX	wcb_high	0.4426	0.4426	1
table_a2	pol_vs_mod	EX	var_low	0.007	0.007	1
table_a2	pol_vs_mod	EX	var_high	0.0095	0.0095	1
table_a2	pol_vs_placebo	PP	ci_lower	-0.6171	-0.6171	1
table_a2	pol_vs_placebo	PP	ci_upper	0.7262	0.7262	1
table_a2	pol_vs_placebo	PP	wcb_low	-0.4339	-0.4339	1
table_a2	pol_vs_placebo	PP	wcb_high	0.5502	0.5502	1
table_a2	pol_vs_placebo	PP	var_low	0.0124	0.0124	1
table_a2	pol_vs_placebo	PP	var_high	0.0114	0.0114	1
table_a2	pol_vs_placebo	EX	ci_lower	-0.6004	-0.6004	1
table_a2	pol_vs_placebo	EX	ci_upper	0.6944	0.6944	1
table_a2	pol_vs_placebo	EX	wcb_low	-0.4491	-0.4491	1
table_a2	pol_vs_placebo	EX	wcb_high	0.535	0.535	1
table_a2	pol_vs_placebo	EX	var_low	0.0085	0.0085	1
table_a2	pol_vs_placebo	EX	var_high	0.0094	0.0094	1
table_a2	mod_vs_placebo	PP	ci_lower	-0.6119	-0.6119	1
table_a2	mod_vs_placebo	PP	ci_upper	0.5044	0.5044	1
table_a2	mod_vs_placebo	PP	wcb_low	-0.4424	-0.4424	1
table_a2	mod_vs_placebo	PP	wcb_high	0.3286	0.3286	1
table_a2	mod_vs_placebo	PP	var_low	0.0106	0.0106	1
table_a2	mod_vs_placebo	PP	var_high	0.0114	0.0114	1
table_a2	mod_vs_placebo	EX	ci_lower	-0.4778	-0.4778	1
table_a2	mod_vs_placebo	EX	ci_upper	0.5769	0.5769	1
table_a2	mod_vs_placebo	EX	wcb_low	-0.3284	-0.3284	1
table_a2	mod_vs_placebo	EX	wcb_high	0.4426	0.4426	1
table_a2	mod_vs_placebo	EX	var_low	0.0083	0.0083	1
table_a2	mod_vs_placebo	EX	var_high	0.0067	0.0067	1
text	pol_vs_mod	PP	no_ds_wcb_low	-1.54	-1.54	1
text	pol_vs_mod	PP	no_ds_wcb_high	1.71	1.71	1
text	pol_vs_mod	PP	no_ds_ci_lower	-1.67	-1.67	1
text	pol_vs_mod	PP	no_ds_ci_upper	1.84	1.84	1
text	pol_vs_mod	PP	ds_wcb_low	-0.34	-0.34	1

Table 3: Ground truth: published value against the value the deposited scripts produce on current R. A blank Archive cell means the script stops before reaching the quantity. *(continued)*

Location	Comparison	DV	Quantity	Paper	Archive	Match
text	pol_vs_mod	PP	ds_wcb_high	0.57	0.57	1
text	pol_vs_mod	PP	ds_ps_wcb_low	-0.34	-0.34	1
text	pol_vs_mod	PP	ds_ps_wcb_high	0.53	0.53	1
text	pol_vs_mod	PP	ds_ps_ci_lower	-0.53	-0.53	1
text	pol_vs_mod	PP	ds_ps_ci_upper	0.7	0.7	1

Of the 130 recorded claims, 128 match the published value and 2 are unreachable, both because of the stale object name.

6 Maintained rewrite

The rewrite lives in `maintained/`: eight scripts covering the three main text tables, the two appendix tables, both figures, and the numbers Sections 4.2 and 4.3 state in prose. It is a translation, not a reanalysis: every estimator, specification and sample restriction is the one the paper used, and the `attrition` functions that implement the method (`estimator_ev`, `estimator_ds`, `sensitivity_ds`, and the internal `ds_manski_cis_2s` the simulation calls) are used unchanged, since they are the paper's contribution rather than incidental tooling.

Table 4: Scripts in the maintained rewrite.

Script	Produces
helpers.R	packages, data, shared row labels
table_1_attrition.R	Table 1: attrition by condition
table_2_outcome_means.R	Table 2: outcome means by condition and response type
table_3_bounds_main.R	Table 3: bounds with and without double sampling
table_a1_mipo_effects.R	Appendix Table 1: effects assuming MIPO
table_a2_pairwise_bounds.R	Appendix Table 2: bounds for all pairwise comparisons
figure_1_simulation.R	Figure 1 and Appendix Figure 1: the simulation study
figure_2_sensitivity.R	Figure 2: the sensitivity analysis
text_section_4_claims.R	the prose numbers in Sections 4.2 and 4.3

Two structural choices are worth stating. `figure_1_simulation.R` writes both Figure 1 and Appendix Figure 1, which differ only in whether the larger sample size is shown, because they come from one simulation grid and splitting them would mean running it twice. `text_section_4_claims.R` reads the CSV output of the table and figure scripts rather than refitting anything, so a number quoted in the text and the same number printed in a table cannot come apart.

6.1 Deprecated patterns replaced

Table 5: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘setwd("")‘	‘here::here()‘
‘read.csv()‘	‘readr::read_csv()‘
‘lm()‘ + ‘commarobust::makerobustseslist()‘	‘estimatr::lm_robust(se_type = "HC2")‘
‘stargazer‘ + ‘out =‘	‘modelsummary(output = ...)‘
‘xtable‘	‘knitr::kable()‘ + ‘readr::write_lines()‘
‘multiplot()‘ (hand-rolled, grid-based)	‘gridExtra::arrangeGrob()‘
triple ‘for‘ loop building a matrix with ‘rbind‘	‘expand_grid()‘ + ‘pmap()‘ + ‘list_rbind()‘
‘reshape2::melt()‘	‘pivot_longer()‘
‘ifelse()‘	‘if_else()‘
‘data.frame()‘	‘tibble()‘
‘scale_color_grey(guide = FALSE)‘	‘scale_color_grey(guide = "none")‘
‘<ggplot> %+% data‘	‘<ggplot> + data‘
magrittr pipe	native pipe

`commarobust::commarobust()` is a wrapper around `lmtest::coeftest(fit, sandwich::vcovHC(fit, type = "HC2"))`, and `estimatr::lm_robust()` defaults to the same HC2 estimator, so the substitution is exact: the standard errors agree to fifteen digits. The one visible difference is in the significance stars. `commarobust::makerobustpslist()` overwrites the intercept's p-value with 1 before handing the models to `stargazer`, which is why the published Appendix Table 1 shows no stars on the constant. The rewrite reports the intercept's actual p-value, so `table_a1_mipo_effects.tex` stars the constant where the published table does not. No estimate or standard error differs.

The simulation deserves a note of its own. The archive passes four objects named `c1a_t`, `c1r_t`, `c2a_t` and `c2r_t` into `ds_manski_cis_2s()` without ever defining them. This does not error, because they map onto formal arguments the function body never touches and R does not evaluate an argument it does not use. They are vestigial cost parameters from an earlier version of the function. The rewrite omits them.

7 Figures

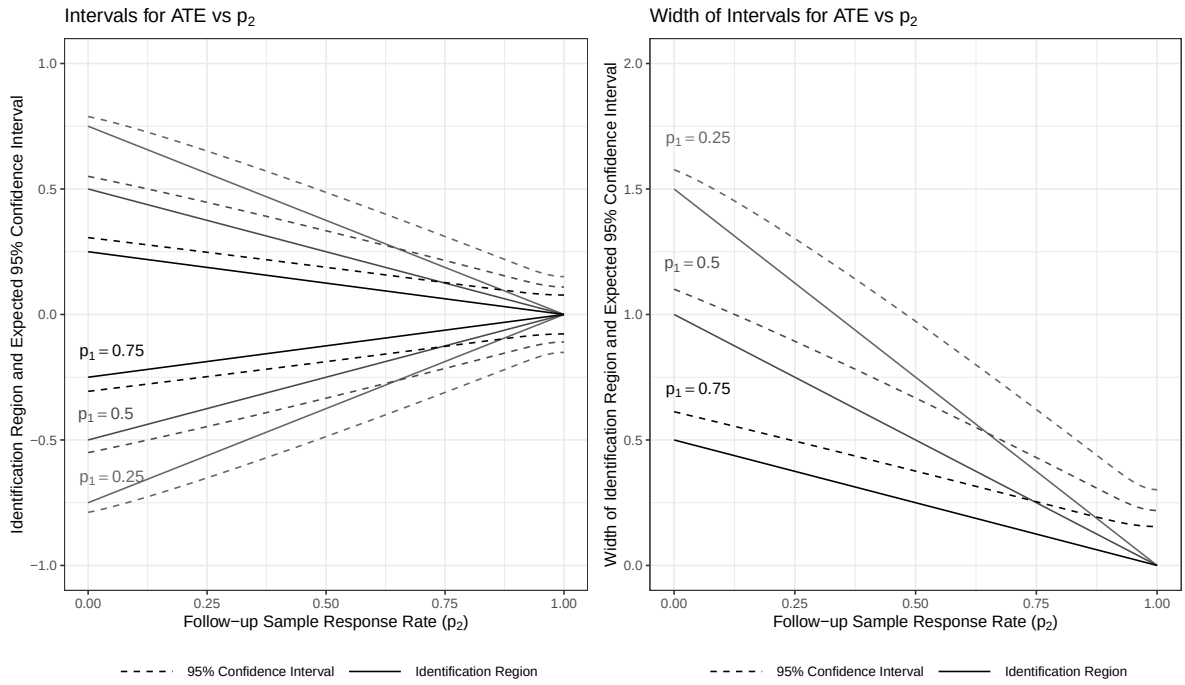


Figure 1: Figure 1 as reproduced by the maintained rewrite: the identification region and expected 95 percent confidence interval for the ATE against the follow-up response rate, at three initial response rates.

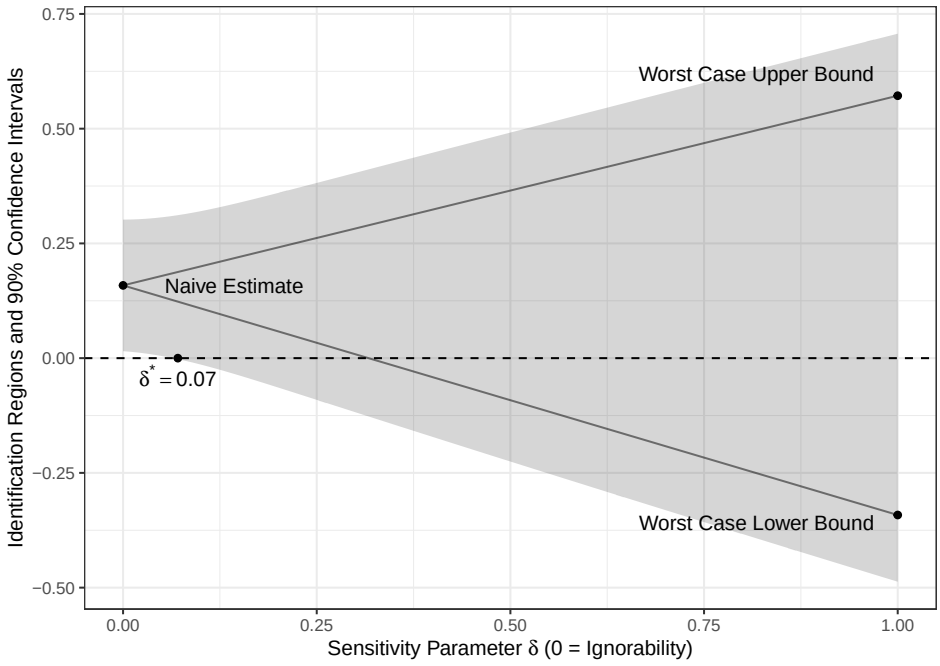


Figure 2: Figure 2 as reproduced by the maintained rewrite: the identification region and 90 percent confidence interval against the sensitivity parameter, with the delta star at which the interval first covers zero.

Neither figure carries printed estimates, so there is no panel label to check the way there is in some archives. Figure 2 is the exception that matters: it is annotated with $\delta^* = 0.07$, and that annotation is the only place the paper states the sensitivity result. The rewrite rebuilds both figures from its own data frames rather than saving the `ggplot` object `sensitivity_ds()` returns, so everything drawn here is produced by code in this repository. `figure_1_simulation.csv` carries the 6,000 point simulation grid behind Figure 1, which is what makes that figure diffable rather than merely inspectable.

8 Maintained rewrite verification

Table 6: Maintained rewrite verification: published value against rewrite output.

Location	Comparison	DV	Quantity	Paper	Rewrite	Match
table_1	Moderate		N_initial	995	995	1
table_1	Moderate		responded	731	731	1
table_1	Moderate		did_not_respond	264	264	1
table_1	Moderate		invited	50	50	1
table_1	Moderate		responded_to_invitation	39	39	1
table_1	Polarized		N_initial	985	985	1
table_1	Polarized		responded	713	713	1
table_1	Polarized		did_not_respond	272	272	1

Table 6: Maintained rewrite verification: published value against rewrite output. *(continued)*

Location	Comparison	DV	Quantity	Paper	Rewrite	Match
table_1	Polarized		invited	50	50	1
table_1	Polarized		responded_to_invitation	33	33	1
table_1	Total		N_initial	1980	1980	1
table_1	Total		responded	1444	1444	1
table_1	Total		did_not_respond	536	536	1
table_1	Total		invited	100	100	1
table_1	Total		responded_to_invitation	72	72	1
table_2	Moderate	PP	IR_mean	3.542	3.54206566347469	1
table_2	Moderate	PP	IR_sd	1.243	1.2426600835923	1
table_2	Moderate	PP	IR_N	731	731	1
table_2	Moderate	PP	DS_mean	3.583	3.58333333333333	1
table_2	Moderate	PP	DS_sd	1.367	1.36730301830269	1
table_2	Moderate	PP	DS_N	39	39	1
table_2	Polarized	PP	IR_mean	3.668	3.6679523141655	1
table_2	Polarized	PP	IR_sd	1.27	1.27045080160387	1
table_2	Polarized	PP	IR_N	713	713	1
table_2	Polarized	PP	DS_mean	3.826	3.82575757575758	1
table_2	Polarized	PP	DS_sd	1.313	1.31322130252061	1
table_2	Polarized	PP	DS_N	33	33	1
table_3	pol_vs_mod	PP	no_ds_ci_lower	-1.6691	-1.66907903866295	1
table_3	pol_vs_mod	PP	no_ds_ci_upper	1.8359	1.83588114535676	1
table_3	pol_vs_mod	PP	no_ds_wcb_low	-1.5391	-1.53914496339566	1
table_3	pol_vs_mod	PP	no_ds_wcb_high	1.7097	1.70966762747749	1
table_3	pol_vs_mod	PP	no_ds_var_low	0.0062	0.0062401008346893	1
table_3	pol_vs_mod	PP	no_ds_var_high	0.0059	0.00588785669626298	1
table_3	pol_vs_mod	PP	ds_ci_lower	-0.5283	-0.528309673828183	1
table_3	pol_vs_mod	PP	ds_ci_upper	0.7452	0.745174826173985	1
table_3	pol_vs_mod	PP	ds_wcb_low	-0.3417	-0.34174537662934	1
table_3	pol_vs_mod	PP	ds_wcb_high	0.5718	0.571815728388134	1
table_3	pol_vs_mod	PP	ds_var_low	0.0129	0.0128647858310974	1
table_3	pol_vs_mod	PP	ds_var_high	0.0111	0.0111080739914738	1
table_3	pol_vs_mod	PP	ds_ps_ci_lower	-0.529	-0.529010551043845	1
table_3	pol_vs_mod	PP	ds_ps_ci_upper	0.6966	0.696600306767075	1
table_3	pol_vs_mod	PP	ds_ps_wcb_low	-0.3444	-0.344389910863888	1
table_3	pol_vs_mod	PP	ds_ps_wcb_high	0.5257	0.525683688852529	1
table_3	pol_vs_mod	PP	ds_ps_var_low	0.0126	0.0125981273119329	1
table_3	pol_vs_mod	PP	ds_ps_var_high	0.0108	0.0107972726598501	1
text	pol_vs_mod	PP	naive_ate_coef	0.126	0.125886650690805	1
text	pol_vs_mod	PP	naive_ate_se	0.066	0.0661527954807814	1
text	pol_vs_mod	PP	naive_ate_pval	0.057	0.0571779950234009	1
text	pol_vs_mod	PP	ci_width_no_ds	3.5	3.50496018401971	1
text	pol_vs_mod	PP	ci_width_ds_ps	1.23	1.22561085781092	1
text	pol_vs_mod	PP	reduction_pct	0.65	0.650321032632869	1
text	pol_vs_mod	PP	delta_star	0.07	0.0707070707070707	1
table_a1	Placebo	PP	wave1_coef	0.352	0.352378065806937	1
table_a1	Placebo	PP	wave1_se	0.067	0.0674342678150233	1
table_a1	Placebo	PP	wave2_coef	0.042	0.0415017522556453	1
table_a1	Placebo	PP	wave2_se	0.066	0.0656684123731188	1
table_a1	Placebo	EX	wave1_coef	0.019	0.0190980494627956	1
table_a1	Placebo	EX	wave1_se	0.035	0.0352828650176962	1
table_a1	Placebo	EX	wave2_coef	-0.011	-0.0105817411887674	1
table_a1	Placebo	EX	wave2_se	0.034	0.0343376300527566	1
table_a1	Polarized	PP	wave1_coef	0.468	0.468375086098891	1
table_a1	Polarized	PP	wave1_se	0.067	0.0665244501080027	1
table_a1	Polarized	PP	wave2_coef	0.126	0.125886650690805	1
table_a1	Polarized	PP	wave2_se	0.066	0.0661527954807814	1
table_a1	Polarized	EX	wave1_coef	-0.031	-0.0311131171539686	1
table_a1	Polarized	EX	wave1_se	0.034	0.034340859741032	1
table_a1	Polarized	EX	wave2_coef	-0.03	-0.0296645452923334	1
table_a1	Polarized	EX	wave2_se	0.034	0.03402111975226	1
table_a1	Constant	PP	wave1_coef	3.108	3.10841313269493	1
table_a1	Constant	PP	wave1_se	0.046	0.0464622897843693	1
table_a1	Constant	PP	wave2_coef	3.542	3.54206566347469	1
table_a1	Constant	PP	wave2_se	0.046	0.0459614434759561	1
table_a1	Constant	EX	wave1_coef	1.531	1.53146374829001	1
table_a1	Constant	EX	wave1_se	0.024	0.0243596218569334	1
table_a1	Constant	EX	wave2_coef	1.617	1.61662106703146	1

Table 6: Maintained rewrite verification: published value against rewrite output. *(continued)*

Location	Comparison	DV	Quantity	Paper	Rewrite	Match
table_a1	Constant	EX	wave2_se	0.024	0.0241507965686312	1
table_a1	all	PP	wave1_N	2156	2156	1
table_a1	all	PP	wave2_N	2156	2156	1
table_a1	all	EX	wave1_N	2156	2156	1
table_a1	all	EX	wave2_N	2156	2156	1
table_a1	all	PP	wave1_R2	0.024	0.0239507001082496	1
table_a1	all	PP	wave2_R2	0.002	0.00174424183477073	1
table_a1	all	EX	wave1_R2	0.001	0.000969193379921962	1
table_a1	all	EX	wave2_R2	4e-04	0.000359806687155073	1
table_a2	pol_vs_mod	PP	ci_lower	-0.5283	-0.528309673828183	1
table_a2	pol_vs_mod	PP	ci_upper	0.7452	0.745174826173985	1
table_a2	pol_vs_mod	PP	wcb_low	-0.3417	-0.34174537662934	1
table_a2	pol_vs_mod	PP	wcb_high	0.5718	0.571815728388134	1
table_a2	pol_vs_mod	PP	var_low	0.0129	0.0128647858310974	1
table_a2	pol_vs_mod	PP	var_high	0.0111	0.0111080739914738	1
table_a2	pol_vs_mod	EX	ci_lower	-0.6087	-0.608731542807715	1
table_a2	pol_vs_mod	EX	ci_upper	0.6032	0.603194110336677	1
table_a2	pol_vs_mod	EX	wcb_low	-0.471	-0.470967068846772	1
table_a2	pol_vs_mod	EX	wcb_high	0.4426	0.442594036170701	1
table_a2	pol_vs_mod	EX	var_low	0.007	0.00701487544541756	1
table_a2	pol_vs_mod	EX	var_high	0.0095	0.00953316194586098	1
table_a2	pol_vs_placebo	PP	ci_lower	-0.6171	-0.617118472435792	1
table_a2	pol_vs_placebo	PP	ci_upper	0.7262	0.726201233938733	1
table_a2	pol_vs_placebo	PP	wcb_low	-0.4339	-0.433926513080827	1
table_a2	pol_vs_placebo	PP	wcb_high	0.5502	0.550203436157751	1
table_a2	pol_vs_placebo	PP	var_low	0.0124	0.012403901338986	1
table_a2	pol_vs_placebo	PP	var_high	0.0114	0.0114487996375541	1
table_a2	pol_vs_placebo	EX	ci_lower	-0.6004	-0.600425131380988	1
table_a2	pol_vs_placebo	EX	ci_upper	0.6944	0.69438536025314	1
table_a2	pol_vs_placebo	EX	wcb_low	-0.4491	-0.449120786151243	1
table_a2	pol_vs_placebo	EX	wcb_high	0.535	0.535009163087335	1
table_a2	pol_vs_placebo	EX	var_low	0.0085	0.00846151811732802	1
table_a2	pol_vs_placebo	EX	var_high	0.0094	0.00938841779116446	1
table_a2	mod_vs_placebo	PP	ci_lower	-0.6119	-0.611862038277648	1
table_a2	mod_vs_placebo	PP	ci_upper	0.5044	0.504351636505354	1
table_a2	mod_vs_placebo	PP	wcb_low	-0.4424	-0.442412292230382	1
table_a2	mod_vs_placebo	PP	wcb_high	0.3286	0.328618863548512	1
table_a2	mod_vs_placebo	PP	var_low	0.0106	0.0106127353002281	1
table_a2	mod_vs_placebo	PP	var_high	0.0114	0.0114143454384199	1
table_a2	mod_vs_placebo	EX	ci_lower	-0.4778	-0.477842686955451	1
table_a2	mod_vs_placebo	EX	ci_upper	0.5769	0.576929833172395	1
table_a2	mod_vs_placebo	EX	wcb_low	-0.3284	-0.328384873083366	1
table_a2	mod_vs_placebo	EX	wcb_high	0.4426	0.442646282695529	1
table_a2	mod_vs_placebo	EX	var_low	0.0083	0.00825624817098751	1
table_a2	mod_vs_placebo	EX	var_high	0.0067	0.00666486134438053	1
text	pol_vs_mod	PP	no_ds_wcb_low	-1.54	-1.53914496339566	1
text	pol_vs_mod	PP	no_ds_wcb_high	1.71	1.70966762747749	1
text	pol_vs_mod	PP	no_ds_ci_lower	-1.67	-1.66907903866295	1
text	pol_vs_mod	PP	no_ds_ci_upper	1.84	1.83588114535676	1
text	pol_vs_mod	PP	ds_wcb_low	-0.34	-0.34174537662934	1
text	pol_vs_mod	PP	ds_wcb_high	0.57	0.571815728388134	1
text	pol_vs_mod	PP	ds_ps_wcb_low	-0.34	-0.344389910863888	1
text	pol_vs_mod	PP	ds_ps_wcb_high	0.53	0.525683688852529	1
text	pol_vs_mod	PP	ds_ps_ci_lower	-0.53	-0.529010551043845	1
text	pol_vs_mod	PP	ds_ps_ci_upper	0.7	0.696600306767075	1

Every row carries `match_rewrite = 1`: **130** values produced by the maintained rewrite, all matching the published paper to reported precision. Every one of them is read out of a file in `maintained/output/` rather than transcribed, so a value in this table that disagreed with the pipeline would be a failure of the pipeline and not of the transcription.

9 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23
Date run	2026-08-01

Table 8: Package versions used for the run behind this report.

Package	Version
attrition	0.1.0
estimatr	1.0.6
modelsummary	2.6.0
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2
purrr	1.2.2
readr	2.2.0
here	1.0.2
gridExtra	2.3
knitr	1.51
kableExtra	1.4.0

Table 9: The deposited files, as pinned in the manifest. All seven served checksums match the checksums Dataverse publishes, and all seven match the local copy this work was done against.

File	Bytes	MD5 (served)	Published MD5 agrees
README.txt	2090	ea327d0299d36ebd0f5bc1b4a01e5c6d	yes
attrition_0.0.0.9000.tar.gz	34511	7fc29b91569bff874735191147a65fc8	yes
commarobust_0.1.0.tar.gz	8085	feef406b22d3259c23d16f91ffbaadc9	yes
experimental_analysis_appendix.R	3756	5fc6e7d1fe25133820d96648ddb7ebfb	yes
experimental_analysis_main_text.R	3668	19a3cf55e8685f65d23419cd013fae0e	yes
levendusky_mturk_clean.csv	178043	6290d4802c5331dfa550fc954ccf07d0	yes
simulation_analysis.R	6358	ebdc8595bddfdef33ee8dde128366039	yes